Teaching Through the Hinge

Equations as Bridges, Constraints and Placeholders in Physics

Paper II of IV

Lee McCulloch-James

Department of Natural and Unnatural Sciences, *Dwight School London*,

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Abstract

Many school equations organise reasoning while leaving part of the physics unspecified. $F_{\text{net}} = ma$ requires an interaction law; $v = f\lambda$ requires a medium model; $P = \Delta E/\Delta t$ requires an energy-transfer pathway. This essay calls such relations *hinges*. A hinge lesson places several physical cases around one shared structure and asks students to identify the additional relation, system boundary, condition or approximation that completes each case. The method supports model selection, transfer and precise interpretation across IB Physics. This is essay II of IV.

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The hinge principle

$\boxed{\text{shared relation}} + \boxed{\text{case-specific closure}} + \boxed{\text{system and assumptions}} \longrightarrow \boxed{\text{physical account}}$

The hinge principle

Physics courses commonly attach one equation to one topic and one familiar diagram. This supports routine calculation, yet students can retain the context while missing the transferable structure. A falling object becomes a "gravity problem", a resistor an "electricity problem", and a diffraction grating a "waves problem". Expert reasoning groups problems more strongly by governing principle and representation [4, 6].

Paper I provides the theoretical justification for this pedagogical pattern: the hinge relation typically occupies a structural or dynamical position in the theory's architecture, while the case-specific closure occupies a constitutive or interactional position [1].

A *hinge* is a compact relation that links parts of a model while leaving a meaningful slot to be filled. Its symbols can recur across mechanisms:

$$F_{\text{net}} = ma, \qquad \tau_{\text{net}} = I\alpha, \qquad P = \frac{\Delta E}{\Delta t}, \qquad v = f\lambda.$$

The missing physics may be an interaction law, constitutive relation, medium property, boundary condition, conservation constraint or approximation. This additional statement supplies *closure*.

For example, $F_{\text{net}} = ma$ describes how net force relates to acceleration. Gravity, elastic deformation and magnetic interaction provide different expressions for the force. The shared equation coordinates the dynamical argument; the force law gives the symbol F its provenance in that case.

The term *hinge* describes a pedagogical role. Each equation retains its established status as a definition, dynamical law, conservation statement, constitutive relation, approximation or derived result. Accurate naming is part of the lesson.

Why teach through hinges?

Model selection

Formula practice often rewards recognition of surface cues. Hinge teaching makes the modelling decision visible. Students identify the shared relation, decide what completes it, declare the system and state the conditions under which the completed model applies. This moves attention from equation recall towards principled categorisation [7].

Transfer

Transfer depends on separating invariant structure from variable context. Contrasting cases make this separation perceptible. Work on contrasting cases shows gains in structural understanding and transfer when comparison precedes or accompanies formal explanation [9, 3]. A hinge set therefore uses cases with a genuine common relation and clearly different physical closures.

System boundaries and causal accounts

Hinge questions repeatedly ask what lies inside the system, what crosses its boundary and what acts externally. These distinctions control the use of momentum conservation, work, power, pressure and induction. They also sharpen causal language. An equation such as $v = f\lambda$ relates wave quantities; the medium relation determines the speed. Charge conservation constrains a nuclear equation; the decay interaction and energetics determine the process.

Connected retrieval

A hinge organises memory around relationships. The prompt "What completes this relation here?" retrieves several linked ideas: the governing structure, the local law, the system, the assumptions and the observable consequence. Such organisation resembles the structured knowledge associated with successful physics problem solving [8].

A hinge map for IB Physics

Table 1: Recurring hinge structures across IB Physics.

Hinge	**Shared relation**	**Cases**	**Closure or condition**
Dynamics	$F_{\text{net}} = ma$	gravity; spring; magnetic force	interaction law and chosen system
Momentum	$F = \Delta p/\Delta t$	collision; rocket; gas pressure	impulse interval, mass flow or collision model
Rotation	$\tau_{\text{net}} = I\alpha$	gravity; friction; magnetic torque	force, lever arm and angular geometry
Power	$P = \Delta E/\Delta t$	lifting; resistor; lamp	transfer pathway, component law and efficiency
Waves	$v = f\lambda$	string; sound; light in a medium	medium-dependent wave speed
Nuclear change	charge and nucleon ledgers	alpha; beta; decay chain	decay mode, energetics and half-life
Surface quantities	total/effective area	pressure; irradiance; magnetic flux	quantity distributed and surface orientation
Number density	number/space	grating; charge carriers; ideal gas	diffraction, transport or gas-law relation
Work	$\Delta E = W$	syringe; gravitational equipotentials; induction	force–displacement relation, potential difference or induced emf

Work functions somewhat differently from the other hinges because it is a process quantity rather than a state relation. The invariant statement is the energy transfer $\Delta E = W$; what varies is the mechanism by which work is evaluated, whether through force and displacement, pressure–volume change, gravitational potential difference or induced electromotive force.
The laws retain their distinct meanings and roles with its value lying in recurrent questions:

- What does the shared relation organise?
- Which statement completes this case?
- Which system, direction, boundary or controlled variable matters?
- Which assumptions limit the result?

Designing a hinge lesson

A compact hinge lesson can follow six moves.

State the shared relation

Present the equation with definitions, units and status. State whether it is a definition, law, conservation relation or approximation. Give only the conditions needed at this stage.

Expose the open slot

Ask what the relation leaves undetermined. Examples include the origin of F, the value of a wave speed, the pathway represented by power, or the process permitted by a conservation ledger.

Compare three cases

Use cases that preserve the shared structure while changing the mechanism. Three cases usually reveal the pattern without overloading the page. Each case should require a separate diagram, boundary statement or physical relation.

Name the closure

Students label the completing statement accurately: interaction law, constitutive relation, equation of state, geometry, conservation condition or approximation. The label helps them understand what sort of knowledge they are using.

Interrogate signs and conditions

Require a statement about direction, sign convention, controlled variable and domain of validity. For example, $P = I^2R$ and $P = V^2/R$ produce different comparisons because one holds current fixed and the other voltage fixed.

Synthesize and transfer

Finish with a comparison table or short paragraph containing three elements:

invariant structure | varying physical closure | resulting observable.

Then give a fourth case with fewer cues. The transfer item tests whether students can reconstruct the architecture.

A useful student prompt

For each case, write one sentence in this form:

The shared relation tells us __________. while the case-specific relation supplies __________.

Assessment

Hinge assessment should credit structure as well as calculation. Suitable prompts include:

- identify the shared relation across three situations;
- supply the missing closure for each case;
- explain why two algebraically equivalent forms support different comparisons;
- draw the system boundary and identify what crosses it;
- classify each statement by epistemic role;
- predict which quantity remains fixed at an interface;
- construct a fourth example governed by the same hinge.

A short marking scheme can allocate credit to four features:

Shared structure	Correct relation, variables and conditions.
Physical closure	Correct case-specific law or constraint.
System & assumptions	Boundary, direction, controlled variable and approximation.
Interpretation	Clear statement of what follows physically.

This format exposes common partial understandings. A student may manipulate the shared equation correctly while choosing an unsuitable closure, or may know the local law while misidentifying the system.

Worked marking example: a partially correct response

Prompt. A charged particle enters a uniform magnetic field perpendicular to its velocity. Use the force hinge to explain why its path is circular and state the assumptions required.

Student response. "The shared relation is $F = ma$. The magnetic force is $F = qvB$, so $qvB = mv^2/r$ and therefore $r = mv/qB$. The force points towards the centre, so the particle travels in a circle."

Shared structure	Correctly uses the dynamical relation and the radial form $a = v^2/r$.	1/1
Physical closure	Correctly supplies $F = qvB$ for perpendicular motion.	1/1
System & assumptions	Omits that the field is uniform, the particle is treated classically, and other forces and radiation losses are neglected.	0/1
Interpretation	Correctly links the inward force to circular motion, though $\lvert q\rvert$ should appear in the radius.	1/1
Total	A structurally sound explanation with incomplete assumptions.	3/4

Cognitive load and representation

The approach reduces load when the invariant relation is held steady and the cases are presented in a common visual grammar and increases load when too many equations, symbols or diagrams appear simultaneously. The design should therefore sequence attention:

1. one shared relation;
2. one case at a time;
3. one explicit comparison;
4. one transfer case.

Worked examples remain useful for unfamiliar procedures. Hinge comparison then consolidates the procedures into a connected schema. Diagrams, equations and prose should carry complementary information and sit close together on the page, consistent with principles for coordinating multiple representations [2]. Cognitive-load theory also supports removing decorative explanation and redundant restatement [10].

Companion hinge worksheets. The classroom worksheets accompanying this essay are available as follows: Energy, Flux, Force, Momentum, Number Density, Power, Torque, and Waves.

Each worksheet places one shared mathematical relation across several physical contexts, asking students to identify the additional law, model or assumption that gives the relation its specific physical meaning.

Safeguards

- **Preserve epistemic status.** Definitions, conservation laws, dynamical laws and constitutive relations require distinct labels.
- **Use genuine structural matches.** Shared notation or similar dimensions alone are insufficient.
- **Keep the syllabus boundary visible.** Extensions should be marked and omitted when they distract from the assessed model.
- **Retain procedural fluency.** Students still need practice in rearrangement, substitution, units, signs and uncertainty.
- **Limit the closure vocabulary.** Introduce technical labels only where they clarify the physics.
- **Choose representative cases.** Each case should add a distinct mechanism, boundary or misconception.

The approach aligns with modelling instruction through its emphasis on systems, representations, assumptions and model deployment [5]. Its narrower contribution is curricular: it identifies equations that can serve as recurring joints between topics.

Teacher planning template

Shared relation	Which equation or conservation structure recurs?
Status	Definition, law, constraint, approximation or derived result?
Open slot	What remains unspecified?
Three cases	Which mechanisms provide a useful contrast?
Closure	Which relation completes each case?
System and conditions	Boundary, direction, controlled variable and assumptions?
Likely conflation	Which quantities may students treat as interchangeable?
Synthesis	How will students state the invariant and variation?
Transfer	Which fourth case will test independent recognition?

Conclusion

Physics gains power through abstraction. Symbols such as F, P, τ, v and n travel across contexts because they suppress local detail. Students need access to both levels: the portable relation and the physical statement that completes it.

Teaching through the hinge makes this architecture explicit. Students compare cases, identify closure, declare systems and conditions, and carry the shared structure into an unfamiliar context. We will broaden our definition of hinge as a shared relation that links layers of of a physical context and its models in order to collate disparate notions under one banner. These are outlined in the appendix and the knowledge graph to the hinge worksheets respects multiple classifications:

Knowledge Graph of hinges

The recurring question across the hinges is always:

What does this equation organise, and what physics completes the case?

Appendix Definition and taxonomy of hinges

A *hinge* is a shared relation that links either different layers of a physical model or different physical contexts. Its subtype depends on the role played by the relation: closing an otherwise incomplete model, constraining a balance, translating between representations, or exposing a common mathematical or geometrical structure.

Closure hinge. A general relation contains a quantity whose physical value must be supplied by a case-specific interaction law or constitutive relation.

Constitutive or response hinge. A system responds to a driving quantity through a material or component parameter, usually within a restricted regime of validity.

Constraint hinge. A conservation law or balance relation constrains transfers across a boundary, junction, or collection of objects.

Representation hinge. One physical process is expressed at several mathematical levels, such as interaction law, differential equation, general solution, and initial conditions.

Structural-form hinge. Different physical systems share the same mathematical form while the variables, parameters, and mechanisms retain different physical meanings.

Geometric hinge. A common geometrical operation, such as spherical spreading, projection, or path difference, generates parallel relations across different physical settings.

Bookkeeping or kinematic hinge. A relation coordinates measured quantities or rates without itself supplying the mechanism that determines them.

Source–field–response hinge. A source law determines a field at a location, and the field then determines the response of a test object.

The table overleaf maps the hinges to types

Worksheet	Hinge type	Shared architecture
Placeholder force	Closure hinge	A general dynamical relation, $F_{\text{net}} = ma$, is completed by gravitational, elastic, electric, magnetic, or frictional force laws.
Torque	Closure hinge	The rotational relation, $\tau_{\text{net}} = I\alpha$, receives its physical content from gravitational, frictional, or magnetic torque.
Power	Closure bookkeeping hinge	Power is a rate of energy transfer, while mechanical, electrical, and radiative laws determine the transfer pathway.
Wave speed	Bookkeeping closure hinge	The relation $v = f\lambda$ coordinates the wave variables; a medium law determines v.
Momentum	Constraint flux hinge	Momentum change is organised through force, impulse, collision balance, rocket momentum flux, and pressure from particle impacts.
Charge conservation	Constraint hinge	Charge and nucleon number restrict allowable nuclear transformations and decay sequences.
Flux	Geometric representation hinge	Force per area, power per area, and magnetic flux share an area-based representation while describing different physical quantities.
Number density	Structural translation hinge	A count per unit length or volume becomes physically predictive only after entering diffraction, current, or ideal-gas relations.
Work	Transfer bookkeeping hinge	Work identifies energy transferred through compression, gravitational motion, or electromagnetic induction.
Boundary balance	Constraint hinge	Input minus output equals accumulation across atmospheric, electrical, and thermal systems.
Field strength	Source–field response hinge	A source produces a field, and the field determines the force on a test mass, charge, or moving charge.
Exponential change	Structural-form hinge	Radioactive decay, capacitor discharge, and attenuation share an exponential form with different characteristic constants and mechanisms.
Potential energy	Configuration representation hinge	The interaction law, configuration variable, and chosen zero determine the potential-energy expression.
Equation of motion	Representation hinge	A physical law generates a differential equation; its solution describes a family of motions selected by initial conditions.
Reciprocal power	Structural-form hinge	Potentials vary as $1/r$, while field strengths and flux densities vary as $1/r^2$ under spherical symmetry.
Refraction	Mechanism-to-geometry hinge	A change in medium changes wave speed and wavelength; oblique incidence then produces a change in direction.
Linear response	Constitutive response hinge	Ohm's law, Hooke's law, and linear stress–strain behaviour share proportional response within a restricted regime.
Rate equations	Bookkeeping hinge	Velocity, current, activity, and power are all expressed as changes of a quantity per unit time.
Interference conditions	Geometric structural hinge	A shared phase or path-difference condition is realised through different source and boundary geometries.

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